

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Indicative content: 1. Set the height of the ramp to about 10cm (above the desk) 2. Measure a distance of 50cm from the end of the ramp and mark this point. 3. Release the squash ball from the top of the ramp starting the stopwatch as you do. 4. When the squash ball reaches the bottom of the ramp press the lap button on the stopwatch. 5. Stop the stopwatch when the squash ball reaches the 50cm mark. Candidate may say there are 2 people using 2 stopwatches. 6. Record the time taken for the ball to travel down the ramp (lap time) and the time to travel 50 cm. 7. To reduce uncertainty could: • repeat experiment (twice more)/compare with different groups • use longer ramp • increase the 50 cm distance • reduce angle of ramp • roll from same point on ramp • same person timing • record the experiment/light gate 5-6 marks Detailed description including both time measurements, sensible distances and how to reduce uncertainty. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		6

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					AO1	AO2	AO3	Total	Maths	Prac
				3-4 marks Appropriate method and some uncertainties <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Limited method and/or uncertainties only <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		0.5/2 subs (1) 0.25 m/s (1)	1	1		2	2	2
		(ii)		0.25 (ecf) /2.5 Subs ecf (1) = 0.1 m/s ² (1)	1	1		2	2	2
				Question 4 total	8	2	0	10	4	10